Five problems
▪ Edge inference is slow due to a heavyweight model
▪ Everything is being transmitted, whether it's useful or not
▪ Security is effectively non-existent
▪ The system degrades badly under poor connectivity
▪ The overall architecture is fragile and lab-only

Chose the transmission filter because it sits at the intersection of the two most immediate operational problems (undifferentiated transmission and poor-link degradation) and it can be built and validated independently of hardware. Everything else (model compression, security, link management) is either infrastructure work or a hardware procurement question, which lies outside of my expertise.

Python file implements a novelty-based transmission filter that sits between the edge classifier and the 5G uplink. It decides which detections are worth transmitting.

Core logic comprises of three components with equal weighting for a novelty score between 0 and 1:

- **Frequency novelty:** what fraction of the detected signal's bandwidth occupies bins not seen in the last 5-minute baseline window? A new frequency band scores 1.0; a well-known one scores 0.
- **Power deviation:** how many standard deviations does the current power reading sit from the rolling mean for that frequency? A sudden 27 dB spike on a known channel is anomalous and gets flagged.
- **Recency:** how long since this frequency was last observed? A sigmoid shaped decay rewards signals that reappear after a gap, as a frequency check would miss burst or intermittent emitters.

Detections above a configurable threshold (default 0.55) are queued for transmission. A separate 'emergency' priority label list also transmits (radar, emergency beacon, satellite downlink) bypasses scoring entirely.

The transmit queue is a fixed-depth deque (default 50). Under a poor link, the oldest items are dropped automatically. The operator can call drain_queue() when the link recovers, batching everything that accumulated.

Key assumptions and trade-offs

The novelty threshold is the main variable which is tuneable: too high and you miss things, too low and you're back to transmitting everything. The filter trades completeness for relevance, which is the right call for situational awareness but means rare legitimate signals can be dropped if they look routine. Security is intentionally lightweight: mTLS covers the transmission channel but doesn't address physical device compromise, which is out of scope for this prototype stage.

Next steps

The most valuable immediate step is threshold calibration i.e. run the filter against a labelled capture to find where the novelty score cuts off genuine signals of interest versus noise. Second could be a comms resilience test: simulate a degraded 5G link and verify queue behaviour matches expectations under real backpressure conditions.

Demo output from the scenario in the code:

In the demo run, the filter transmits roughly 5 of 16 events (~31%), with the rest dropped as routine.

```
── Transmission filter demo ──────────────────────────────
     t    freq MHz    power  label                   score  decision
──────────────────────────────────────────────────────────────────────────────
     0s     935.200    -72 dBm  gsm_downlink           1.000  novelty
     2s     935.200    -71 dBm  gsm_downlink           0.575  novelty
     4s     935.200    -73 dBm  gsm_downlink           0.575  novelty
     6s     935.200    -72 dBm  gsm_downlink           0.241  dropped
     8s     935.200    -71 dBm  gsm_downlink           0.477  dropped
     3s    2412.000    -65 dBm  wifi_beacon            1.000  novelty
     5s    2412.000    -66 dBm  wifi_beacon            0.575  novelty
     7s    2412.000    -65 dBm  wifi_beacon            0.408  dropped
    10s    1090.000    -55 dBm  ads_b                  1.000  novelty
    11s     935.200    -72 dBm  gsm_downlink           0.288  dropped
    13s     935.200    -45 dBm  gsm_downlink           0.575  novelty
    15s     406.000    -80 dBm  emergency_beacon       1.000  priority_label
    17s     162.025    -60 dBm  vhf_marine             1.000  novelty
    20s    2412.000    -65 dBm  wifi_beacon            0.385  dropped
    22s    2412.000    -85 dBm  unknown                0.575  novelty
    28s     935.200    -72 dBm  gsm_downlink           0.354  dropped

── Filter stats ──────────────────────────────────────────
  seen                          16
  queued                        10
  dropped                       6
  priority                      1
  queue_depth                   10
  transmission_rate_pct         62.5

── Queued for transmission (10 detections) ────────────────
  gsm_downlink            935.200 MHz  score=1.000  reason=novelty
  gsm_downlink            935.200 MHz  score=0.575  reason=novelty
  gsm_downlink            935.200 MHz  score=0.575  reason=novelty
  wifi_beacon             2412.000 MHz  score=1.000  reason=novelty
  wifi_beacon             2412.000 MHz  score=0.575  reason=novelty
  ads_b                   1090.000 MHz  score=1.000  reason=novelty
  gsm_downlink            935.200 MHz  score=0.575  reason=novelty
  emergency_beacon        406.000 MHz  score=1.000  reason=priority_label
  vhf_marine              162.025 MHz  score=1.000  reason=novelty
  unknown                 2412.000 MHz  score=0.575  reason=novelty
```